

Installation and Configuration of phpMyAdmin with MariaDB

Objective: To install and configure phpMyAdmin with MariaDB on RHEL 9 and verify database management through a web browser.

1. Check the Change Request: Verify if the change is assigned and check its status (Approved/Pending). Start work only after approval.

2. Login: Get the server IP from the TL and log in using SSH, RDP, VNC, or Remote Console.

3.Take Backup: Always take a backup before changes. Use NetBackup for physical servers or snapshot for virtual machines.

4.Configure Network and Hostname

Set network adapter to NAT (automatic) and set hostname as –

hostnamectl set-hostname phpmyadminapp.radical.com

```
[root@dnsserver ~]# hostnamectl set-hostname phpmyadminapp.radical.com
[root@dnsserver ~]# hostname
phpmyadminapp.radical.com
[root@dnsserver ~]#
[root@dnsserver ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.117.136 netmask 255.255.255.0 broadcast 192.168.117.255
    inet6 fe80::20c:29ff:fe16:33 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:16:00:33 txqueuelen 1000 (Ethernet)
    RX packets 833 bytes 868741 (848.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 512 bytes 38413 (37.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 83 bytes 9238 (9.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 83 bytes 9238 (9.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

5.Configure Repository (Verify and Install Required Packages)

#yum search httpd

#yum install httpd.x86_64

#yum install mariadb-server

#yum install php php-mysqld php-json php-mbstring php-xml php-cli

Note- Package Requirement:

httpd is required to provide web services, mariadb-server is required for database management, and PHP packages (php, php-mysqlnd, php-json, php-mbstring, php-xml, php-cli) are required to run the phpMyAdmin web application and establish communication with the MariaDB database server.

#yum search httpd

```
[root@dnsserver ~]# yum search httpd
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:03:47 ago on Tue 16 Jun 2026 02:25:19 PM IST.
===== Name Exactly Matched: httpd =====
httpd.x86_64 : Apache HTTP Server
===== Name & Summary Matched: httpd =====
httpd-core.x86_64 : httpd minimal core
```

#yum install -y httpd.x86_64

```
[root@dnsserver ~]# yum install -y httpd.x86_64
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:04:53 ago on Tue 16 Jun 2026 02:25:19 PM IST.
Dependencies resolved.
=====
Package                Arch      Version           Repository        Size
=====
Installing:
httpd                   x86_64    2.4.53-11.el9_2.4 AppStream         54 k
Installing dependencies:
apr                     x86_64    1.7.0-11.el9      AppStream         127 k
apr-util                x86_64    1.6.1-20.el9      AppStream         98 k
apr-util-bdb            x86_64    1.6.1-20.el9      AppStream         15 k
=====
```

#systemctl enable --now httpd

Purpose: Starts and enables the Apache web server.

#yum search mariadb-server

```
[root@dnsserver ~]# yum search mariadb-server
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:06:08 ago on Tue 16 Jun 2026 02:25:19 PM IST.
===== Name Exactly Matched: mariadb-server =====
mariadb-server.x86_64 : The MariaDB server and related files
```

```
#yum install mariadb-server.x86_64 -y
```

```
[root@dnsserver ~]# yum install mariadb-server.x86_64 -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-
manager to register.

Last metadata expiration check: 0:07:20 ago on Tue 16 Jun 2026 02:25:19 PM IST.
Dependencies resolved.
=====
Package                        Arch      Version                               Repository    Size
=====
Installing:
mariadb-server                 x86_64    3:10.5.16-2.el9_0                   AppStream     9.4 M
Installing dependencies:
mariadb                       x86_64    3:10.5.16-2.el9_0                   AppStream     1.6 M
mariadb-common                 x86_64    3:10.5.16-2.el9_0                   AppStream      39 k
=====
```

Purpose: Installs the MariaDB database server package.

Note: ****PhpMyAdmin** is built with **PHP**, so we need to install **PHP** and some required extensions using the following command**. PHP is required to run the phpMyAdmin web application. The packages php-mysqldb, php-json, php-mbstring, php-xml, and php-cli provide database connectivity and additional functionality required by phpMyAdmin.

```
# yum install php php-mysqldb php-json php-mbstring php-xml php-cli -y
```

```
[root@phpmyadminapp ~]#
[root@phpmyadminapp ~]# yum install php php-mysqldb php-json php-mbstring php-xml php-cli -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:30:04 ago on Tue 16 Jun 2026 02:25:19 PM IST.
Dependencies resolved.
=====
Package                        Architecture    Version                               Repository    Size
=====
Installing:
php                            x86_64          8.0.27-1.el9_1                       AppStream     11 k
php-cli                        x86_64          8.0.27-1.el9_1                       AppStream     3.1 M
php-common                     x86_64          8.0.27-1.el9_1                       AppStream     686 k
php-mbstring                   x86_64          8.0.27-1.el9_1                       AppStream     473 k
php-mysqldb                    x86_64          8.0.27-1.el9_1                       AppStream     154 k
php-xml                         x86_64          8.0.27-1.el9_1                       AppStream     138 k
Installing dependencies:
nginx-filesystem               noarch          1:1.20.1-14.el9                       AppStream     13 k
php-ndb                       x86_64          8.0.27-1.el9_1                       AppStream     87 k
=====
```

Enable mariadb service

```
#systemctl enable --now mariadb
```

```
[root@phpmyadminapp ~]# systemctl enable --now mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@phpmyadminapp ~]#
```

6. Secure MariaDB

mysql_secure_installation

```
[root@phpmyadminapp ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y

Remove anonymous users? [Y/n] y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
```

Root password was configured successfully and MariaDB installation was secured.

Purpose: Secures the MariaDB installation by setting the root password, removing anonymous users, removing the test database, and disabling remote root login.

7. Login to MariaDB and create a database radical

#mysql -u root -p (Enter the root password configured in the previous step.)

#create database radical;

#show databases;

```
[root@phpmyadminapp ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.5.16-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database radical;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| radical |
+-----+
4 rows in set (0.001 sec)
```

Purpose: Access the MariaDB command-line interface and Create a database that will be managed through phpMyAdmin.

8. Create Database User

```
#CREATE USER 'admin'@'localhost' IDENTIFIED BY 'redhat';
```

Grant Privileges

```
#GRANT ALL PRIVILEGES ON radical.* TO 'admin'@'localhost';
#FLUSH PRIVILEGES;
```

```
MariaDB [(none)]> CREATE USER 'admin'@'localhost' IDENTIFIED BY 'redhat';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON radical.* TO 'admin'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

Purpose: Creates a dedicated user account for database access and Grants the user full access to the radical database.

****Verify User Login****

- Exit MariaDB:
#exit;
- Login using the newly created user:
#mysql -u admin -p
- Verify:
#show databases;

```

MariaDB [(none)]> exit
Bye
[root@phpmyadminapp ~]# mysql -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.16-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database                |
+-----+
| information_schema       |
| radical                  |
+-----+
2 rows in set (0.001 sec)

```

Purpose: Confirms that the user authentication and permissions are working correctly.

9. Download phpMyAdmin

```
#cd /tmp
```

```
#wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz
```

```

[root@phpmyadminapp tmp]#
[root@phpmyadminapp tmp]# wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz
--2026-06-16 15:35:35-- https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz
Resolving www.phpmyadmin.net (www.phpmyadmin.net)... 79.127.235.11, 79.127.235.52, 79.127.235.9, ...
Connecting to www.phpmyadmin.net (www.phpmyadmin.net)|79.127.235.11|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://files.phpmyadmin.net/phpMyAdmin/5.2.3/phpMyAdmin-5.2.3-all-languages.tar.gz [following]
--2026-06-16 15:35:36-- https://files.phpmyadmin.net/phpMyAdmin/5.2.3/phpMyAdmin-5.2.3-all-languages.tar.gz
Resolving files.phpmyadmin.net (files.phpmyadmin.net)... 79.127.235.6, 79.127.235.12, 79.127.213.245, ...
Connecting to files.phpmyadmin.net (files.phpmyadmin.net)|79.127.235.6|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14446385 (14M) [application/octet-stream]
Saving to: 'phpMyAdmin-latest-all-languages.tar.gz'

phpMyAdmin-latest-all-lan 100%[=====] 13.78M  3.12MB/s   in 4.4s

2026-06-16 15:35:41 (3.12 MB/s) - 'phpMyAdmin-latest-all-languages.tar.gz' saved [14446385/14446385]

```

***Purpose:** Downloads the latest phpmyadmin package*

Extract phpMyAdmin

```
#tar -xzf phpmyadmin-latest-all-languages.tar.gz
```

```
#ls
```

```

[root@phpmyadminapp tmp]# tar -xzf phpMyAdmin-latest-all-languages.tar.gz
[root@phpmyadminapp tmp]# ls
phpMyAdmin-5.2.3-all-languages  phpMyAdmin-latest-all-languages.tar.gz
[root@phpmyadminapp tmp]#

```

Move phpMyAdmin to Web Directory

```
#mv phpmyadmin-5.2.3-all-languages /var/www/html/phpmyadmin
```

```
#ls /var/www/html/phpmyadmin
```

```
[root@phpmyadminapp tmp]# mv phpMyAdmin-5.2.3-all-languages /var/www/html/phpmyadmin
[root@phpmyadminapp tmp]#
[root@phpmyadminapp tmp]#
[root@phpmyadminapp tmp]# ls /var/www/html/phpmyadmin
babel.config.json      CONTRIBUTING.md      js                   README              sql                 yarn.lock
Changelog              doc                  libraries            RELEASE-DATE-5.2.3  templates
composer.json          examples             LICENSE              robots.txt          themes
composer.lock          favicon.ico          locale               setup                url.php
config.sample.inc.php  index.php            package.json         show_config_errors.php vendor
[root@phpmyadminapp tmp]#
```

Purpose: Moves the extracted phpmyadmin application files to Apache's document root so they can be accessed through a web browser.

10. Configure phpMyAdmin

Create Configuration File

```
# cd /var/www/html/phpmyadmin
```

Create configuration file:

```
#cp config.sample.inc.php config.inc.php
```

```
[root@phpmyadminapp tmp]# cd /var/www/html/phpmyadmin
[root@phpmyadminapp phpmyadmin]# cp config.sample.inc.php config.inc.php
[root@phpmyadminapp phpmyadmin]#
```

Purpose: Creates the main phpmyadmin configuration file from the sample template.

** Generate Blowfish Secret

Generate a random secret key:

```
# openssl rand -base64 32
```

```
[root@phpmyadminapp phpmyadmin]# openssl rand -base64 32
LAK44uZqCENliNWzfxrJ/B8BBDcneXpRkKAiC48IfXg=
[root@phpmyadminapp phpmyadmin]#
```

Purpose: Generates a secret key used by phpmyadmin for secure cookie authentication. (phpMyAdmin requires a secret key for cookie authentication.)

Edit Configuration File

```
#vi config.inc.php
```

Locate:

```
→$cfg['blowfish_secret'] = '';
```

Modify it:

```
→$cfg['blowfish_secret'] = 'generated_secret_key';
```

```
[root@phpmyadminapp phpmyadmin]# vi config.inc.php
[root@phpmyadminapp phpmyadmin]#
```

```
<?php
/**
 * phpMyAdmin sample configuration, you can use it as base for
 * manual configuration. For easier setup you can use setup/
 *
 * All directives are explained in documentation in the doc/ folder
 * or at <https://docs.phpmyadmin.net/>.
 */

declare(strict_types=1);

/**
 * This is needed for cookie based authentication to encrypt the cookie.
 * Needs to be a 32-bytes long string of random bytes. See FAQ 2.10.
 */
$config['blowfish_secret'] = 'Lak44uZqCENliNWzfxrJ/B8BBDcNeXpRkKAiC48IfXg='; /* YOU MUST FILL IN THIS FOR COOKI
E AUTH! */

/**
 * Servers configuration
 */
```

Purpose: Configures secure cookie-based authentication for phpMyAdmin.

Apply SELinux context:

```
# restorecon -Rv /var/www/html/phpmyadmin
```

```
[root@phpmyadminapp phpmyadmin]# restorecon -Rv /var/www/html/phpmyadmin
```

Purpose: Restores the default SELinux security contexts on phpmyadmin files and directories, allowing Apache (httpd) to access them correctly.

11. Restart and enable the services :

```
#systemctl restart httpd
```

```
#systemctl enable --now httpd
```

```
[root@phpmyadminapp phpmyadmin]# systemctl restart httpd
[root@phpmyadminapp phpmyadmin]#
[root@phpmyadminapp phpmyadmin]#
[root@phpmyadminapp phpmyadmin]# systemctl enable --now httpd
[root@phpmyadminapp phpmyadmin]#
```

```
#systemctl restart mariadb
```

```
#systemctl enable --now mariadb
```

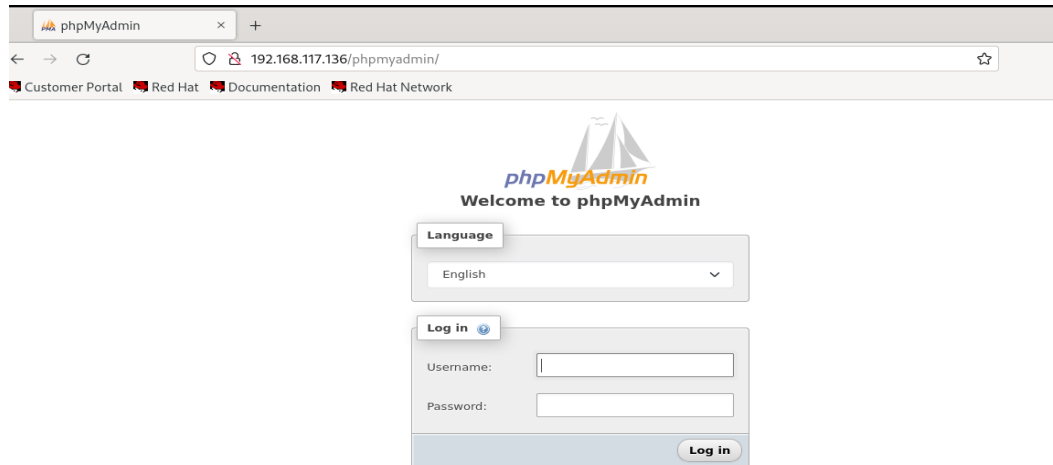


```
[root@phpmyadminapp phpmyadmin]# systemctl restart mariadb.service
[root@phpmyadminapp phpmyadmin]#
[root@phpmyadminapp phpmyadmin]# systemctl enable --now mariadb
[root@phpmyadminapp phpmyadmin]#
```

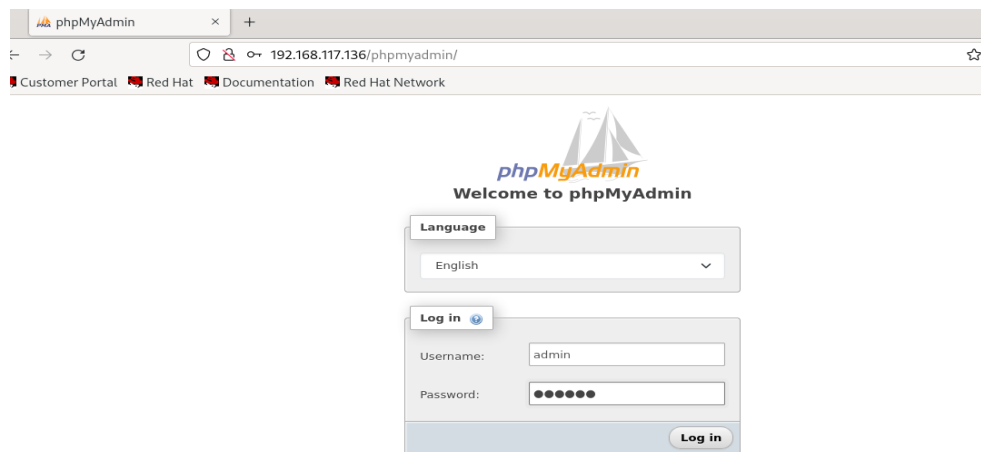
12. Local Testing (Access phpmyadmin) :

Open browser:

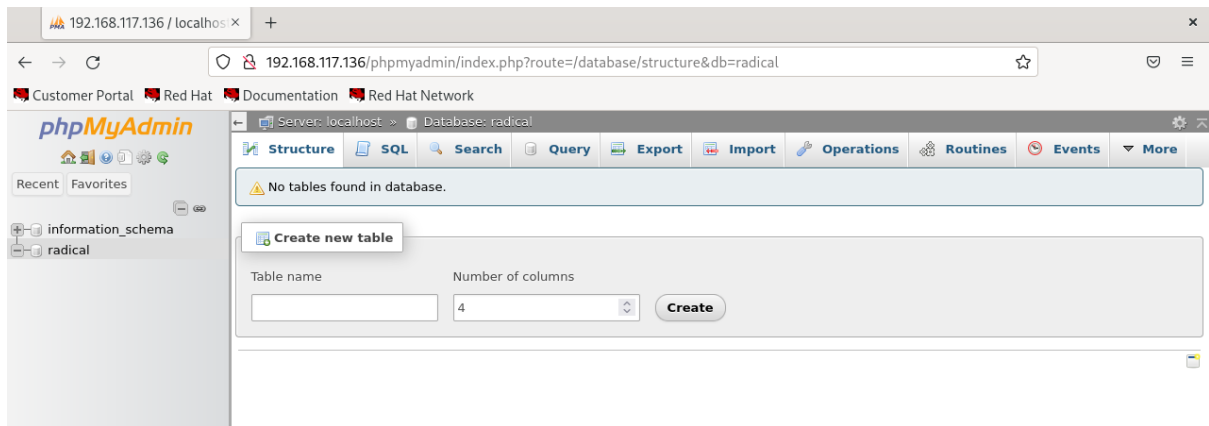
<http://192.168.117.136/phpmyadmin/>



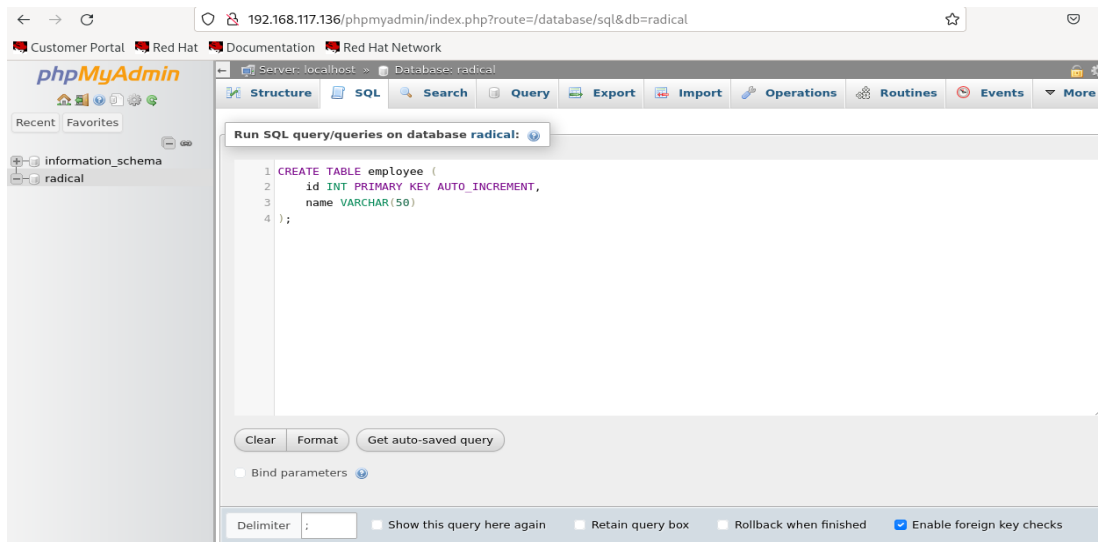
Login via admin –



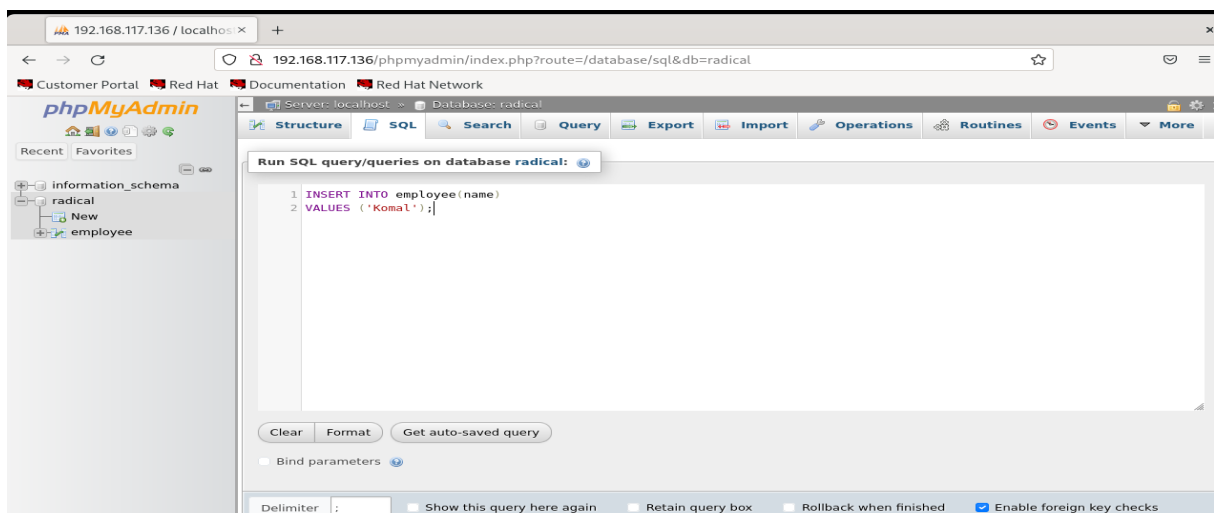
Dashboard of phpmyadmin. We can see the database radical we created earlier.



Create
a table employee:



Insert values into table:



13. Set SELinux temporarily to permissive mode and stop the firewall for testing purposes.

```
#systemctl stop firewalld.service
```

```
#setenforce 0
```

```
root@dnserver:/var/www/html/phpmyadmin

[root@phpmyadminapp phpmyadmin]# systemctl stop firewalld.service
[root@phpmyadminapp phpmyadmin]#
[root@phpmyadminapp phpmyadmin]# setenforce 0
[root@phpmyadminapp phpmyadmin]#
[root@phpmyadminapp phpmyadmin]# getenforce
Permissive
[root@phpmyadminapp phpmyadmin]#
```

14. Remote Testing :

Login to the RHEL 8 client machine, ensure it is on the same network, and verify connectivity by pinging the server IP.

```
root@silcharclient:~/Desktop

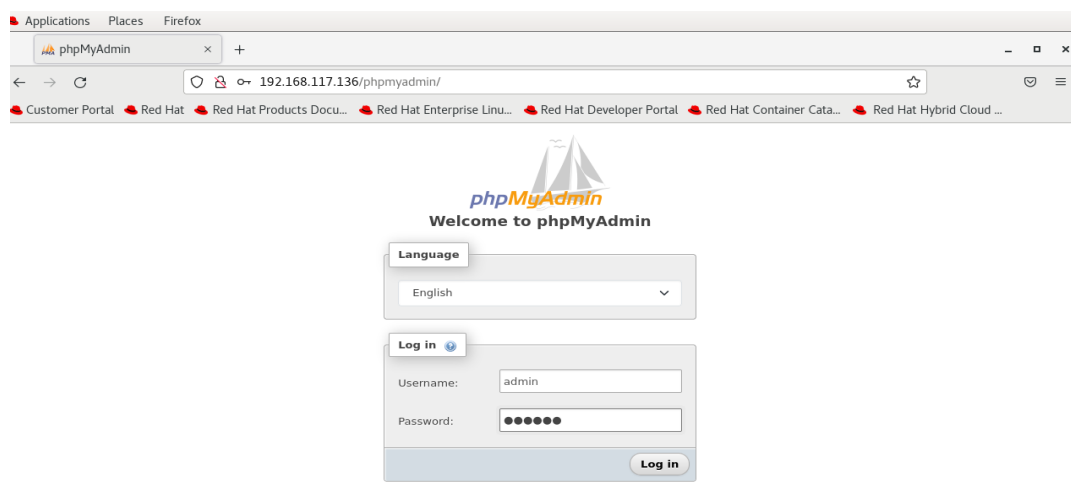
File Edit View Search Terminal Help

[root@silcharclient Desktop]# ifconfig ens160
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.117.133  netmask 255.255.255.0  broadcast 192.168.117.255
    inet6 fe80::20c:29ff:fe1c:298f  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:1c:29:8f  txqueuelen 1000  (Ethernet)
    RX packets 190  bytes 226153 (220.8 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 80  bytes 8046 (7.8 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

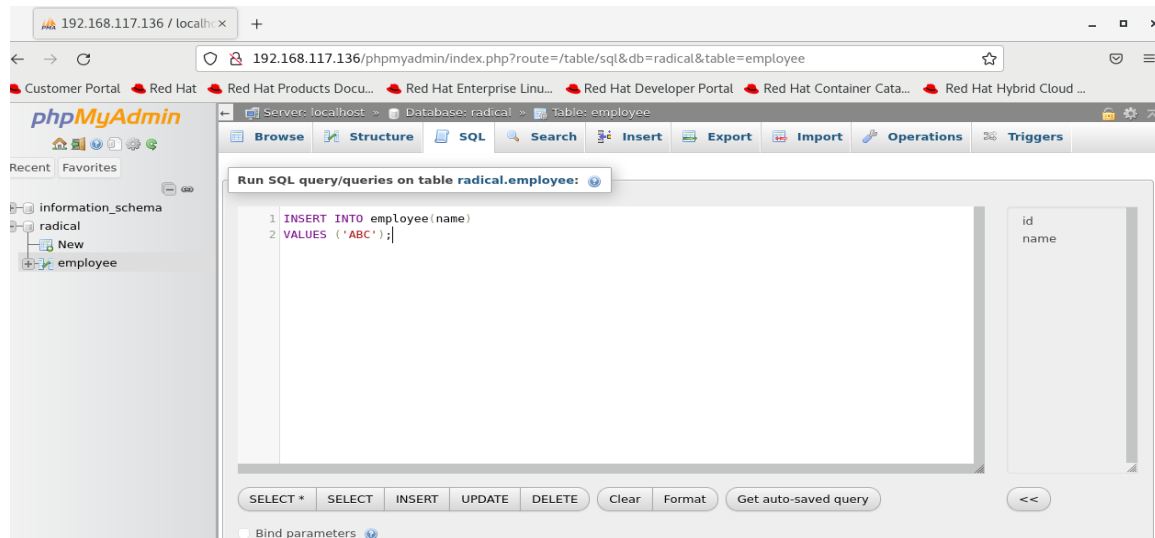
[root@silcharclient Desktop]# ping 192.168.117.136 -c2
PING 192.168.117.136 (192.168.117.136) 56(84) bytes of data.
64 bytes from 192.168.117.136: icmp_seq=1 ttl=64 time=0.628 ms
64 bytes from 192.168.117.136: icmp_seq=2 ttl=64 time=0.794 ms

-- 192.168.117.136 ping statistics --
2 packets transmitted, 2 received, 0% packet loss, time 1014ms
rtt min/avg/max/mdev = 0.628/0.711/0.794/0.083 ms
root@silcharclient Desktop]#
```

Open client's browser – <http://192.168.117.136/phpmyadmin>

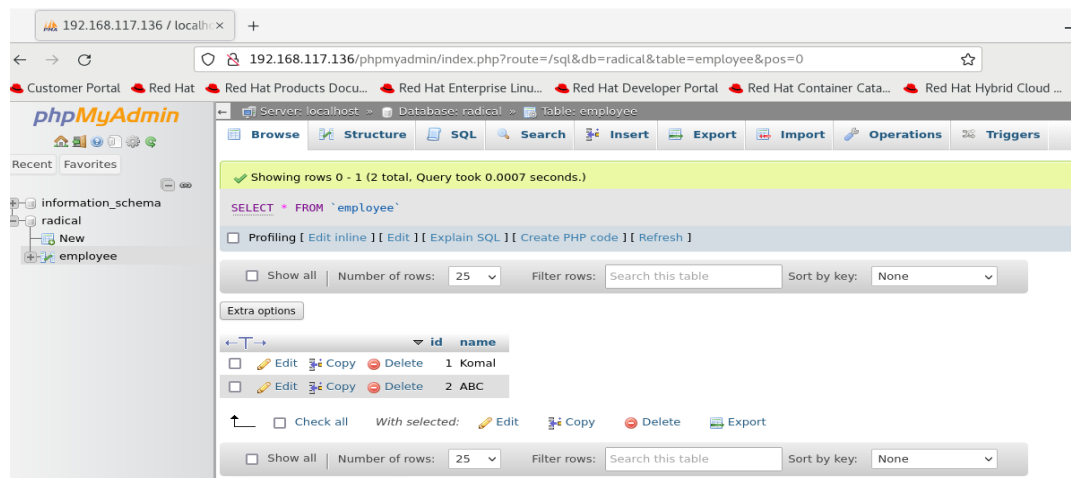


Insert values into employee table from client machine:



Verification:

The employee table was successfully created and records were inserted from both the local server and the remote RHEL 8 client system.



Result: phpMyAdmin is successfully installed and configured with MariaDB on RHEL 9. Database operations such as database creation, user creation, table creation, and record insertion are successfully performed through the phpMyAdmin web interface from both local and remote client systems.

Conclusion:

The objective of installing and configuring phpMyAdmin with MariaDB on RHEL 9 is achieved successfully. phpMyAdmin provided a web-based graphical interface for managing MariaDB databases, simplifying administration and database operations.